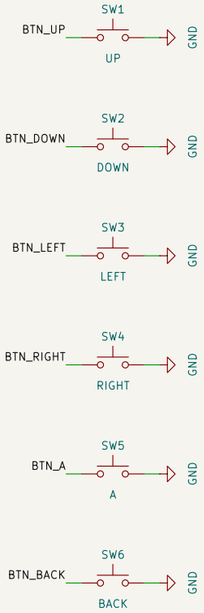
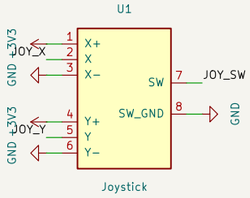


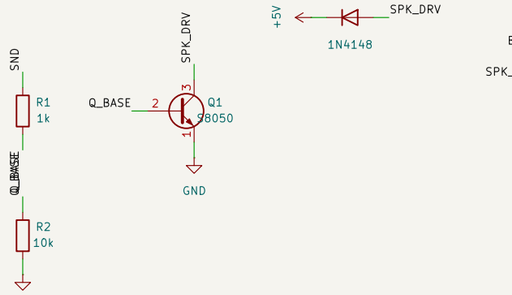

J1 = DEB header 1 (top edge)
odd pins = outer row, pin 1 at New-TFT end
unused pins are left open (X)



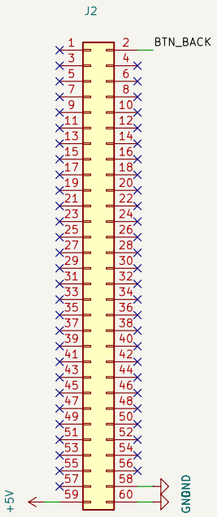

Buttons: active LOW, no resistors —
the STM32 internal pull-ups are enabled
in Input_Init(). BACK (PB8) is in parallel
with the DEB's own KEY1.




Stick pots run from 3V3, NOT 5V
(ADC pins max out at VDDA = 3.3 V).
If an axis is reversed, flip JOY_INVERT_X/Y
in fw_console/App/input.c.




Sound: PA6 (TIM3_CH1 PWM) -> S8050 low-side switch on +5V.
Front: 12 mm PASSIVE buzzer through R3. Back: optional 8 ohm speaker on J3
through R4. Fit either or both. D1 catches the coil kick.
Firmware Sound_Stop() parks PA6 low, so the transistor idles OFF.




J2 = DEB header 2 (bottom edge)
odd pins = inner row, pin 1 at New-TFT end




Power comes from the DEB: its USB 5 V and its AMS1117 3.3 V.
The carrier draws < 0.3 A from 5 V (sound only) and ~1 mA from 3V3.




M3 holes line up with the DEB's corner holes:
11 mm M3 standoffs, DEB to carrier.

Pin map = fw_console/PINMAP.md (PE1-4 d-pad, PC0/PC1/PC4 stick, PC5 A, PB8 BACK, PA6 soun
Carrier for mcudev STM32F4XX_Pro (STM32F407ZGT6) + 3.2in TFT

**NYCU Maker Club**

Sheet: /
File: stm32_console_carrier.kicad_sch

***Title: STM32 Arcade — console carrier***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-10 | **Rev: 0.1** |
| KiCad E.D.A. 10.0.0 | | Id: 1/1 |